

Essential n-3 fatty acids in pregnant women and early visual acuity maturation in term infants.

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BACKGROUND: Docosahexaenoic acid (DHA) is important to neural development. Whether DHA intakes are low enough in some pregnant women to impair infant development is uncertain. OBJECTIVE: We sought to determine whether DHA deficiency occurs in pregnant women and contributes to poor infant development. DESIGN: Biochemical cutoffs, dietary intakes, or developmental scores indicative of DHA deficiency are not defined. Infant development has a distribution in which an individual's potential development is unknown. This was a randomized intervention to establish a distribution of developmental scores for infants of women with DHA intakes considered to be above requirements against which to compare the development of infants of mothers consuming their usual diet. DHA (400 mg/d; n = 67) or a placebo (n = 68) was consumed by the women from 16 wk gestation until delivery. We determined maternal red blood cell ethanolamine phosphoglyceride fatty acids, dietary intakes at 16 and 36 wk gestation, and infant visual acuity at 60 d of age. RESULTS: We described an approach to identify DHA deficiency when biochemical and functional markers of deficiency are unknown. In multivariate analyses, infant visual acuity was related to sex (beta = 0.660, SE = 0.93, and odds ratio = 1.93) and maternal DHA intervention (beta = 1.215, SE = 1.64, and odds ratio = 3.37). More infant girls in the placebo than in the DHA intervention group had a visual acuity below average (P = 0.048). Maternal red blood cell ethanolamine phosphoglyceride docosatetraenoic acid was inversely related to visual acuity in boys (rho = -0.37, P < 0.05) and girls (rho = -0.48, P < 0.01). CONCLUSIONS: These studies suggest that some pregnant women in our study population were DHA-deficient.